

Data Analytics for Personalized Genomics and Precision Medicine

Scribing of Lecture 8 – Classification (Oct 2)

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In previous lecture (L7):

- Prof. Li revised the basic concepts regarding clustering, distance and similarity.
- He explained thoroughly on hierarchical clustering and Mahalanobis distance.

In this lecture (L8):

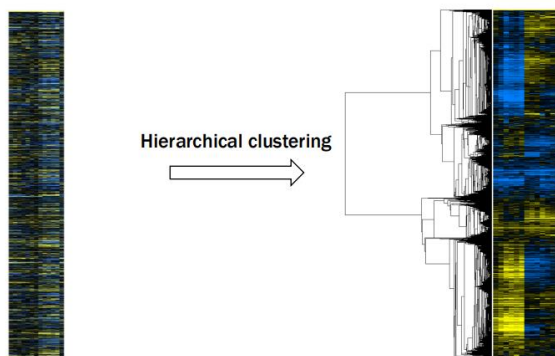
- Prof. Li briefly recapped on the **hierarchical clustering** and **Mahalanobis distance**.
- He focused on explaining the purposes and procedures of **classification** as well as the classification algorithm **K-nearest neighbors**. He also compared the differences between clustering and classification for clarification.

Mahalanobis distance (Partial Recap):

- Motivation: calculate the distance while also considering data distribution.

Hierarchical clustering (Partial Recap):

- Produces a set of nested clusters organized as a hierarchical tree.
 - Example: Initial and clustered data.



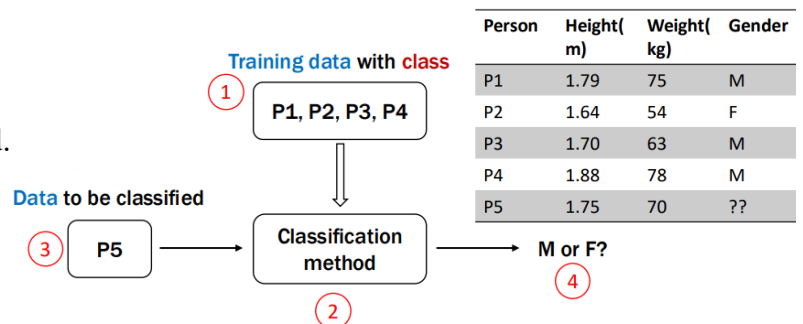
- Steps: **1.** Compute the similarity or distance matrix, **2.** Let each data point be a cluster, **3.** Merge the two closest clusters, **4.** Update the similarity or distance matrix, **5.** Repeat steps 3 and 4 until only a single cluster remains.
- Ways to update distance matrix: minimum, maximum, group average and distance

between centroids.

- Example: refer to previous lecture (L7) scribing.
- Clustering: intra-cluster distances are small while inter-clusters ones are large
 - Use similarity or dissimilarity to measure the difference, e.g. cosine similarity, correlation, Euclidean, Manhattan and Mahalanobis distances.

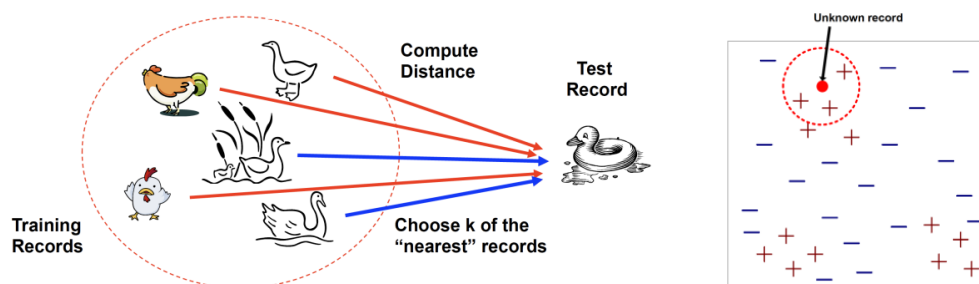
Classification:

- Motivation:
 - For items: achieve better organization and know where to put new items.
 - For people: provide different treatment/strategies for different groups.
 - For biology: classify the profile of a new gene expression as normal or tumor.
- Meaning: given a collection of records (training set), where each record contains a set of **attributes**, and one of the attributes is the **class**, classification is a method to assign the class of **previously unseen records** based on their other attributes and the training set.
- Procedures:
 1. Given training data with class.
 2. Establish the classification method.
 3. Get the data to be classified.
 4. Obtain the classification result.



K-nearest neighbors:

- Analogy: if it walks like a duck, quacks like a duck, then it is probably a duck.

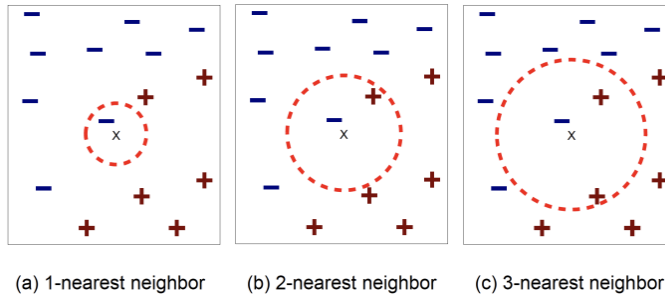


- KNN: a simple algorithm that stores all available instances and classifies new instances based on a **distance or similarity metric** to the available ones
- Assumption and processes:
 - Training process: store the available training instances.
 - Predicting process: find the K training instances that are closest to the query instance and return the most frequent class label among those K instances.

- Data should be normalized.

➤ Things to consider when using KNN:

- A distance or similarity metric, e.g. cosine similarity, correlation, Euclidean, Manhattan and Mahalanobis distances.
- Number of neighbors to look at (K). Example:



- An optional weighting function for neighbors (uniform or distance-based).

➤ In practice: a value of K between 5 and 10 gives good results for **most low-dimensional data sets**. A good K can be chosen by using cross-validation (given in L11).

➤ Procedure: suppose we have chosen the distance metric and K,

- 1. Normalization.
- 2. Compute distances.
- 3. Identify the K most similar data.
- 4. Take their class out and find the mode class.

➤ Example:

- Suppose we have chosen the Euclidean distance and $K = 2$ for the following data:

Person	Height(m)	Weight(kg)	Gender
P1	1.79	75	M
P2	1.64	54	F
P3	1.70	63	M
P4	1.88	78	M
P5	1.75	70	??

- 1. Perform column-wise, min-max normalization to put features on common range.

Person	Height	Weight	Gender
P1	0.625	0.875	M
P2	0	0	F
P3	0.25	0.375	M
P4	1	1	M
P5	0.4583	0.6667	??

- 2. Compute distances (Euclidean or straight-line distances).

Person	P5	Gender
P1	0.267	M
P2	0.809	F
P3	0.358	M
P4	0.636	M
P5	0	??

- 3. Identify the K (chosen as 2) most similar data.

Person	P5	Gender
P1	0.267	M
P2	0.809	F
P3	0.358	M
P4	0.636	M
P5	0	??

- 4. Take their class out and find the mode class.

- Hence, the mode class of P1 and P3 is M, and P5 is classified as Male.

Clustering versus classification:

- Clustering contains:

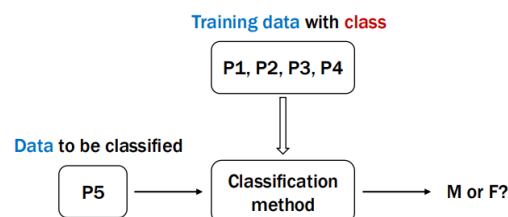
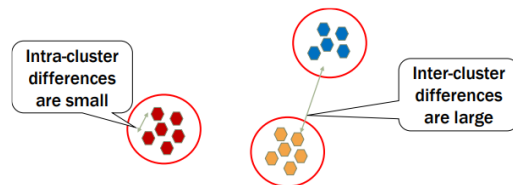
- Data to be clustered
- Similarity measurement
- Clustering algorithm

- Classification contains:

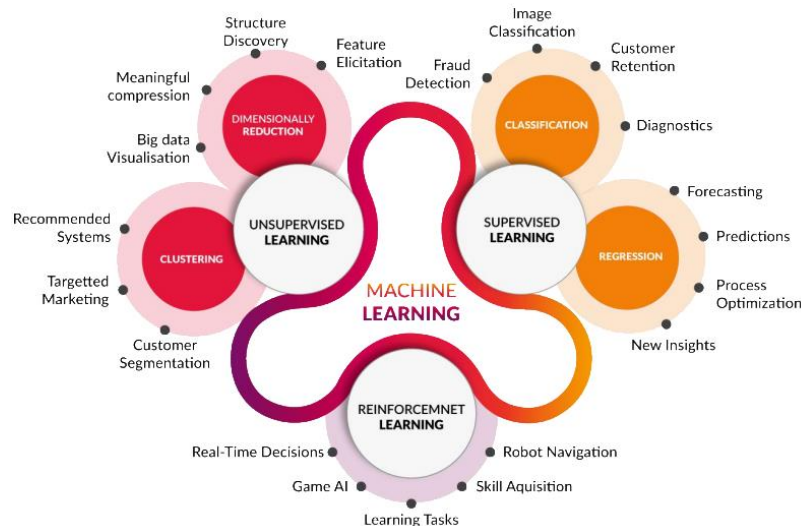
- Data to be classified
- Training data with class
- Classification method

- Comparison table:

	Clustering	Classification
Goal	Find similarity (clusters) in the data	Assign class to the new data
Data	Data without class	Training data with class and testing data without class
Classes	Unknown number of classes	Known number of classes
Output	Cluster index for each point	Class assignment of the testing data
Algorithm	One phase	Two phases (training and application)



- Unsupervised learning: machine learning algorithms to **analyze** and **cluster** unlabeled data, e.g. **clustering** and dimension reduction
- Supervised learning: machine learning algorithms to **classify** and **predict** outcomes, trained on labeled data, e.g. **classification** and regression
- Structure of machine learning:



- Potential project suggested by Prof. Li:
 - Data preprocessing for gene expression matrix:
 - ◆ Data collection and merging
 - ◆ Exploration
 - ◆ Visualization
 - ◆ Data cleaning
 - ◆ **Classification** (instead of clustering)

Takeaway from Prof. Li:

- Understand what is happening under the package and black box.
- Understand what we are coding and achieving.
- Hence, use such packages and tools correctly.

Resources and uncovered topics:

- Introduction to Data Mining: Chapters 4.1, 4.2, and 5.2
- Problem of KNN
- Logistic regression and neural network
- Decision tree, SVM and Bayesian
- Model overfitting

- Cross-validation
- KNN in Python: <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>